berry Pi GPIO nodes. Arrange these in the UI, as shown in Fig. 4.

Make connections between the switches and pins to complete the flow. In switch node, there is a grey bubble on the right. Right-click on it and drag the cursor to the corresponding pin (GPIO) of Raspberry Pi, as shown in Fig. 5. Repeat the process for the other two switch nodes. For LED dimmer application, drag and drop the slider from UI node list and connect it to a pin of Raspberry Pi.

Configuring the nodes

Now, we need to configure the nodes according to hardware connections (Fig. 9). Use two tabs info and debug on the right of the window shown in Fig. 2 to know more about each node. Click on the node and all details about it will be displayed under info. Debug tab is used to monitor the function of the flow.

Deploy button is used as Run, which is used to start the flow.

To configure the node, double-click on it. An empty window will open. Make the following changes (also shown in Fig. 6):

In node properties window, change the following:

Name: Tube Light, Group: Room1, On Value: 1, Off Value: 0

Press Done to save the changes.

Do the same for other switch nodes with different names, as shown in Fig. 8.

To configure pin node, double-click on it from the window shown in Fig. 5.

In rpi gpio properties window (Fig. 7), make the following changes:

Pin: Select the GPIO pin you are using, Type: Digital output, check Initialize pin state and select Initial level of pin – low (0). This will define all the pins initially low (off).

Name: GPIO_xx (which is you are using)

Press Done to save changes.

Do the same for the other two switches, except the slider. Input of the relay driver and LED dimmer circuit should be connected with GPIO pins of Raspberry Pi board, as configured in Node-RED flow. For the slider, define output type as PWM.

Final configured nodes are shown in Fig. 8.

The circuit diagram of the home

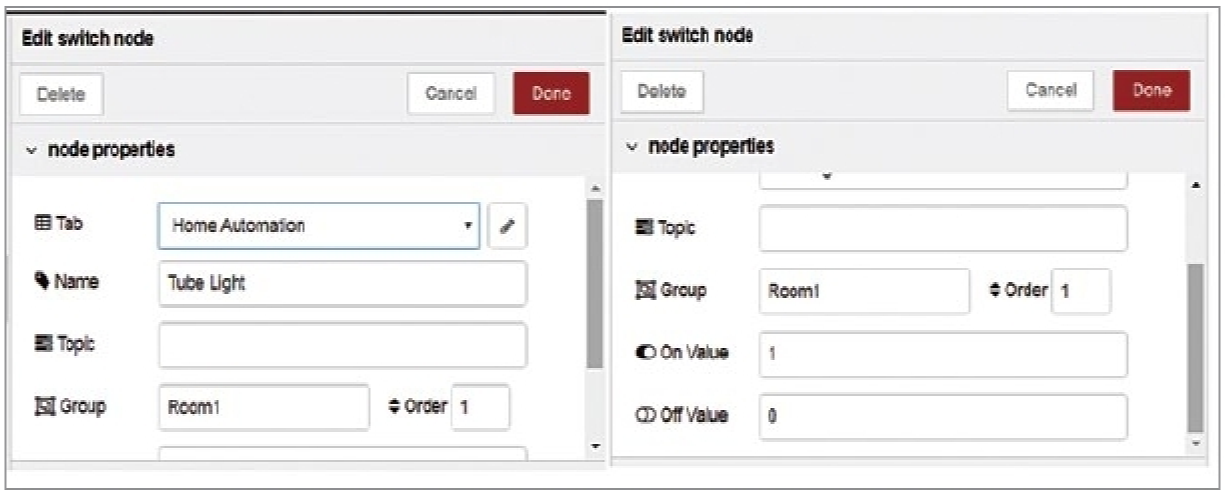

Fig. 6: Editing of switch nodes

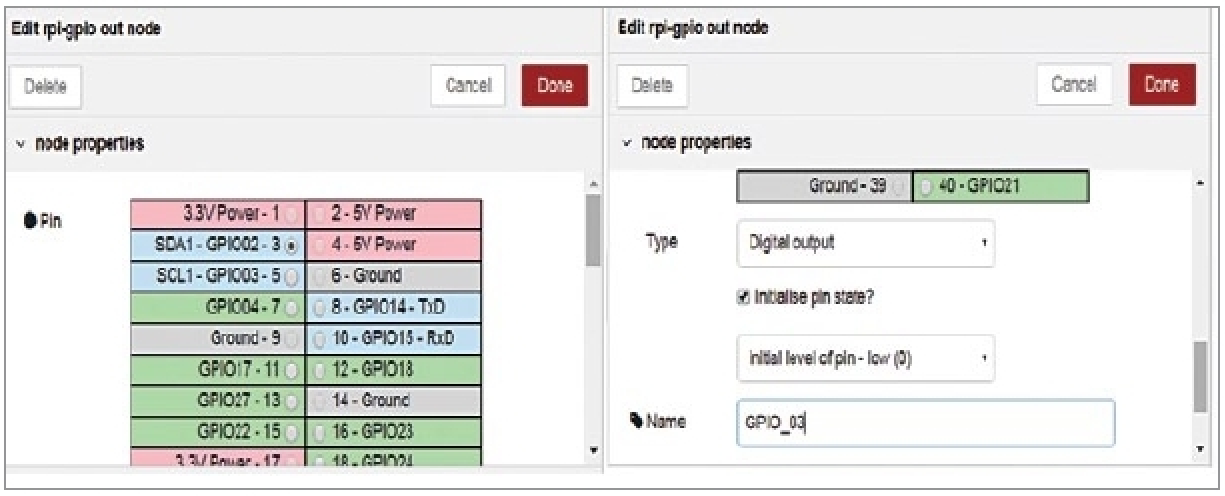

Fig. 7: Defining GPIO pins

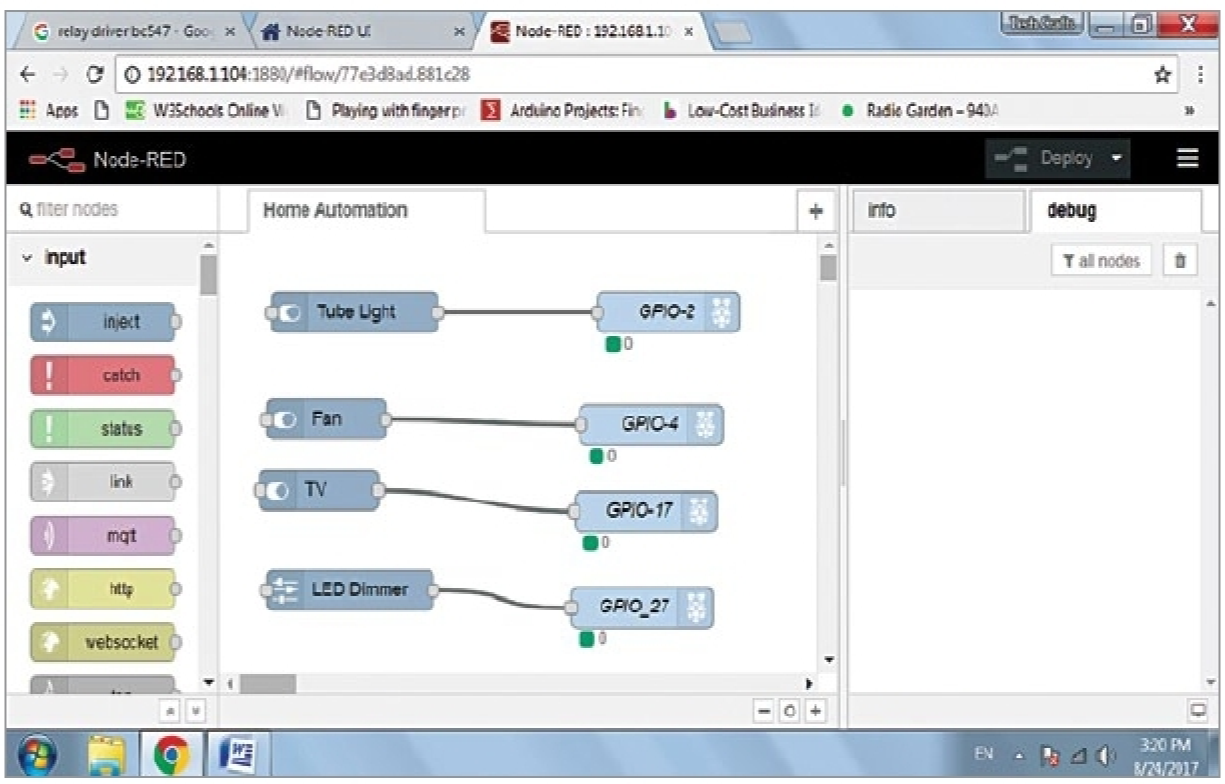

Fig. 8: Representation of nodes

PARTS LIST	
Semiconductors:	
Board1	- Raspberry Pi 3 board
D1-D3	- 1N4007, rectifier diode
LED1-LED4	- 5mm LED
T1-T4	- BC547, npn transistor
Resistors (all 1/4-watt, ±5% carbon):	
R1-R8	- 1-kilo-ohm
Miscellaneous:	
BATT.1	- 12V battery
CON1, CON3, CON5	- 2-pin connectors for load
CON2, CON4, CON6	- 2-pin terminal connectors for 230V AC
RL1-RL3	- 12V single-changeover relay
	- Jumper wires